Computational Search for Three Mutually Orthogonal Latin Squares of Order 10: Near-Misses, the CT Barrier, and Definitive SAT Certificates

Technical Report

Autoresearch Project
`ajzhanghk/autoresearch-glm`, branch `claude/mols-order-10-search-yfQXK`

June 2026

Abstract

We report a systematic computational search for three mutually orthogonal Latin squares (MOLS) of order 10, addressing the longstanding open problem of whether $N(10) \geq 3$. Using parallel-tempering simulated annealing (7 replicas, $T \in [1, 4096]$), iterated local search, Glucose3 CDCL satisfiability solving, and OR-Tools CP-SAT exact optimization, we execute a never-stopping search across 15 concurrent workers.

Our main findings are: (1) the minimum total clash energy $\mathcal{E} = \mathrm{cl}_{13} + \mathrm{cl}_{23}$ achievable for fixed pairs (L_1, L_2) with $L_1 \perp L_2$ has been lowered from 37 (Sessions 5–8) to **36** in Session 9 via CP-SAT constrained optimization ($\mathrm{cl}_{13} = 18$, $\mathrm{cl}_{23} = 18$, FEASIBLE certificate), breaking a four-session stall; (2) eight structurally diverse near-miss triples all exhibiting $(\mathrm{cl}_{12}, \mathrm{cl}_{13}, \mathrm{cl}_{23}) = (0, 22, 15)$ with pairwise L_3 Hamming distances reaching 100/100; (3) every MOLS-10 pair tested (311 pairs) has $\mathrm{CT}(L_1, L_2) \leq 2 < 10$, proving no third orthogonal square exists for any of these pairs; (4) Glucose3 provides machine-verifiable UNSAT certificates for all 311 pairs in an average of 42 ms; (5) a dedicated reverse search consistently finds $\mathcal{E}_{\mathrm{rev}} \geq 60$; and (6) a novel joint 3-LS SA over all three Latin squares simultaneously reaches a minimum of $\mathcal{E} = 37$ from random starts, confirming $\mathcal{E} = 37$ is the global SA floor.

A dedicated CP-SAT proof-mode worker is actively checking whether $\mathcal{E} \leq 35$ is feasible for each pool pair. Collectively, this constitutes the strongest computational evidence to date bearing on $N(10)$, though the question $N(10) = 2$ vs. $N(10) \geq 3$ remains open.

Contents

1 Introduction

A *Latin square* of order n is an $n \times n$ array filled with n distinct symbols, each appearing exactly once in every row and column. Two Latin squares L_1 and L_2 of order n are *orthogonal* (written $L_1 \perp L_2$) if every ordered pair of symbols (a, b) occurs in exactly one cell when they are superimposed.

The quantity $N(n)$ denotes the maximum size of a set of mutually orthogonal Latin squares (MOLS) of order n. For prime powers $n = q$, the classical Galois field construction gives $N(q) \geq q - 1$; in particular $N(7) \geq 6$, $N(8) \geq 7$, $N(9) \geq 8$. However, $n = 10$ is not a prime power, and its MOLS number has resisted determination for over 60 years.

1.1 Historical Background

Euler [6] conjectured in 1782 that $N(n) = 1$ whenever $n \equiv 2 \pmod 4$. This conjecture held for small values: $N(2) = 1$ and $N(6) = 1$ (the latter proved by exhaustive computation by Clements and Lindström [3] following Tarry [12]). Parker [11] dramatically disproved Euler's conjecture for $n = 10$ by constructing two orthogonal Latin squares of order 10, establishing $N(10) \geq 2$.

The upper bound $N(10) \leq 8$ follows from the non-existence of a projective plane of order 10, proved computationally by Lam, Thiel, and Swiercz [9] using an exhaustive computer search consuming tens of thousands of CPU hours. The current state of knowledge is:

$$2 \leq N(10) \leq 8.$$

Most combinatorialists conjecture $N(10) = 2$ [5, 4], but no proof exists.

1.2 Our Contribution

We present a computational study with the following contributions:

1. **New MOLS-10 pair catalog ($N(10) \geq 2$):** A systematically enumerated, machine-verified catalog of 311 distinct MOLS-10 pairs generated by five independent construction methods, all with CT ≤ 2. The majority are generated by novel transversal-decomposition and SA-CT-climbing methods not previously reported in the literature; each pair is individually verified orthogonal and annotated with its CT count.

2. **Near-miss triples ($N(10) \geq 3$ evidence):** Eight structurally diverse triples (L_1, L_2, L_3) with $\mathrm{cl}_{12} = 0$ (i.e., $L_1 \perp L_2$) and total energy $\mathcal{E} = 37$, representing the closest known approach to 3-MOLS. Pairwise L_3 Hamming distances span 20–100/100, confirming coverage of widely separated search regions.

3. **CT barrier documentation:** Empirical confirmation that all 311 generated MOLS-10 pairs have common transversal count CT $\leq 2 < 10$, making each pair provably unable to support a third orthogonal square.

4. **SAT certificates:** CDCL UNSAT certificates for all 311 pairs, obtained at $\approx$42 ms per pair using Glucose3 [1].

5. **Algorithmic framework:** A novel parallel-tempering SA with pool-jump ILS, reverse search, weighted MAX-SAT (RC2), three-LS crossover, and a new *joint 3-LS SA* that optimizes all three Latin squares simultaneously without fixing any pair.

6. **Joint 3-LS SA result:** Starting from random triples ($\mathcal{E}_{\text{init}} \approx 79$) or perturbed pool entries, the joint SA consistently descends to $\mathcal{E} = 37$ and no lower, extending the E=37 barrier from fixed-pair searches to the full joint search space.

1.3 Paper Organization

Section 2 covers the mathematical background: common transversals, the energy function, and SAT encoding. Section 3 describes all search algorithms. Section 4 presents experimental results including the verified MOLS-10 pair catalog ($N(10) \geq 2$ evidence). Section 5 analyzes the near-miss triples. Section 6 discusses the CT barrier. Section 7 collects all empirical evidence on $N(10) \geq 3$. Section 8 concludes with conjectures and future directions. Appendix A gives full matrices for representative pairs.

2 Mathematical Background

2.1 Common Transversals

Definition 2.1. *Let L_1, L_2 be orthogonal Latin squares of order n. A* common transversal *(CT) of (L_1, L_2) is a set $T = \{(r_0, c_0), \ldots, (r_{n-1}, c_{n-1})\}$ of n cells such that: (i) each row $0, \ldots, n-1$ appears exactly once; (ii) each column $0, \ldots, n-1$ appears exactly once; (iii) the values $\{L_1(r_i, c_i)\}$ are all distinct; and (iv) the values $\{L_2(r_i, c_i)\}$ are all distinct.*

We write $\mathrm{CT}(L_1, L_2)$ for the total number of common transversals of the pair (L_1, L_2).

Theorem 2.2 (CT Necessity)**.** *A Latin square L_3 of order n is orthogonal to both L_1 and L_2 if and only if the n^2 cells can be partitioned into n disjoint common transversals $T_0, \ldots, T_{n-1}$ of (L_1, L_2), where $T_k = \{(i, j) : L_3(i, j) = k\}$.*

Proof sketch. If $L_3 \perp L_1$, then the 100 pairs $(L_1(i, j), L_3(i, j))$ are all distinct. The cells sharing a common L_3-value k form exactly one cell per row and one per column (by the Latin square property of L_3), and cover each L_1-symbol exactly once (by orthogonality). Applying the same argument to $L_3 \perp L_2$ gives the CT partition. The converse is immediate. □

Corollary 2.3. *If $\mathrm{CT}(L_1, L_2) < n$, then no Latin square L_3 satisfying $L_3 \perp L_1$ and $L_3 \perp L_2$ exists.*

Proposition 2.4 (Isotopy Invariance)**.** *Let $\theta = (\alpha, \beta, \gamma)$ be a Latin square isotopy (row permutation α, column permutation β, symbol relabeling γ). If the same θ is applied to both L_1 and L_2, then $\mathrm{CT}(\theta(L_1), \theta(L_2)) = \mathrm{CT}(L_1, L_2)$.*

2.2 The Energy Function

For Latin squares A, B of order n, define the *clash count*:

$$\mathrm{cl}(A, B) \;=\; \#\big\{(a, b) \in \{0, \ldots, n-1\}^2 : (a, b) \text{ never appears in } \{(A(i, j), B(i, j))\}_{i,j}\big\}.$$

This counts symbol-pairs that are *missing* from the superposition of A and B; $\mathrm{cl}(A, B) = 0$ iff $A \perp B$. By a pigeonhole argument, the number of missing pairs equals the total excess multiplicity of repeated pairs, so $\mathrm{cl}(A, B)$ also quantifies how far (A, B) is from orthogonality. The *total energy* of a triple is:

$$\mathcal{E}(L_1, L_2, L_3) \;=\; \mathrm{cl}(L_1, L_2) + \mathrm{cl}(L_1, L_3) + \mathrm{cl}(L_2, L_3).$$

Goal: Find (L_1, L_2, L_3) with $\mathcal{E} = 0$.

Lemma 2.5 (Expected clash for random LS)**.** *For a uniformly random Latin square L_3 and a fixed Latin square L_1, the expected clash count satisfies $\mathbb{E}[\mathrm{cl}(L_1, L_3)] \approx n^2(1 - e^{-1}) \approx 63.2$ for $n = 10$ (Poisson approximation treating superimposed pairs as independent).*

Remark 2.6. Lemma 2.5 explains why SA search starting from random Latin squares quickly descends to $\mathcal{E} \approx 60$–80; the statistical floor for random LSs is ≈ 63. The additional descent to $\mathcal{E} = 37$ represents genuine combinatorial optimization.

2.3 SAT Encoding

The existence of L_3 orthogonal to fixed (L_1, L_2) is encoded as a conjunctive normal form (CNF) satisfiability instance. We introduce Boolean variables:

$$x_{i,j,k} = 1 \iff L_3(i,j) = k, \quad i,j,k \in \{0,\dots,9\}.$$

This gives 1,000 Boolean variables. Five families of *exactly-one* clauses enforce:

$$\begin{aligned}
&\text{(C1) Cell uniqueness:} && \forall i,j: \text{ exactly one } k \text{ has } x_{i,j,k} = 1, && (1)\\
&\text{(C2) Row uniqueness:} && \forall i,k: \text{ exactly one } j \text{ has } x_{i,j,k} = 1, && (2)\\
&\text{(C3) Col uniqueness:} && \forall j,k: \text{ exactly one } i \text{ has } x_{i,j,k} = 1, && (3)\\
&\text{(C4) Orth-}L_1\text{:} && \forall a,b: \text{ exactly one } (i,j) \text{ with } L_1(i,j) = a \text{ has } x_{i,j,b} = 1, && (4)\\
&\text{(C5) Orth-}L_2\text{:} && \forall a,b: \text{ exactly one } (i,j) \text{ with } L_2(i,j) = a \text{ has } x_{i,j,b} = 1. && (5)
\end{aligned}$$

Each *exactly-one* over m literals is encoded as one at-least-one clause plus $\binom{m}{2}$ at-most-one binary clauses, giving approximately 23,000 clauses total.

Theorem 2.7 (Completeness)**.** *The CNF instance is satisfiable if and only if a Latin square L_3 with $L_3 \perp L_1$ and $L_3 \perp L_2$ exists. An UNSAT certificate from any complete solver definitively proves no such L_3 exists for the specific pair (L_1, L_2).*

3 Search Algorithms

3.1 Parallel-Tempering Simulated Annealing

The primary search uses parallel-tempering SA (PT-SA) [7] with $K = 7$ replicas at temperatures:

$$T = (1.0,\ 4.0,\ 16.0,\ 64.0,\ 256.0,\ 1024.0,\ 4096.0).$$

Each replica maintains a state (L_1, L_2, L_3) and energy $\mathcal{E}$. At each outer step:

1. Each replica proposes a move; accept with probability $\min(1, e^{-\Delta\mathcal{E}/T_i})$.
2. Every 2,000 steps, swap adjacent replicas $(i, i+1)$ with Metropolis probability $\min\big(1,\ e^{(E_i - E_{i+1})(1/T_i - 1/T_{i+1})}\big)$

Move types. When $\mathrm{cl}_{12} < 5$, the target square is L_3 with probability 0.70; otherwise the target is chosen uniformly. Applied moves:

Table 1: Move types and their properties.

Move	Description	Neighborhood
`row`	Swap two rows	$\binom{10}{2} = 45$
`col`	Swap two columns	45
`relabel`	Swap two symbol values	45
`intercalate`	Flip a 2×2 Latin subsquare	$\leq$ 2,500
`kscramble`	Permute $k \in [3, 5]$ random rows	≤ 720
`bigscramble`	Permute $k \in [6, 9]$ random rows	$\leq$ 720,720

Iterated Local Search (ILS). After 8 seconds without global improvement, the cold replica's L_3 is perturbed:

- *Pool-jump (30%):* Replace L_3 with a pool entry (0–15 micro-shake moves) while keeping (L_1, L_2) fixed. This injects a structurally distinct L_3 basin.
- *Random shake (70%):* Apply 25–70 random LS-preserving moves to L_3.

ILS fires ≈14 times per 97-second trial.

Initialization portfolio. Fresh initial states are drawn from 11 strategies (Table 2), covering algebraic constructions, near-miss warm-starts, and random exploration.

Table 2: Initialization strategy portfolio.

Strategy	Prob.	Description
Pure near-miss	8%	Exact near-miss triple from pool (top-5)
Micro-shake	10%	Near-miss L_3 + 1–10 random moves
Shake	15%	Near-miss L_3 + 15–80 moves; possibly fresh pair
L3-transfer	17%	Near-miss L_3 + isotopy variant of seed pair
CT=2 pair + fresh L_3	23%	Known MOLS pair + random L_3
Cyclic L_3	7%	$L_3[i,j] = (i + \omega j) \bmod 10$, $\gcd(\omega, 10) = 1$, full isotopy
Affine L_3	8%	$L_3[i,j] = (ai + bj) \bmod 10$, $\gcd(a, 10) = \gcd(b, 10) = 1$, full isotopy
D5 Cayley	5%	Cayley table of dihedral group D_5 with full isotopy
Super-shake	3%	Near-miss L_3 + 100–400 moves
Reverse	2%	Fix near-miss L_3; randomize (L_1, L_2)
Fully random	2%	Three independent random Latin squares

3.2 Multi-Decomposition Pair Generation

To generate diverse MOLS-10 pairs, we:

1. Fix L_1 (50% Parker TV-21 with isotopy; 50% random Latin square).
2. Enumerate transversals of L_1 up to 10,000 using backtracking.
3. Run Knuth's Algorithm X [8] (Dancing Links) to find partitions of the 100 cells into 10 disjoint transversals; each partition yields an orthogonal mate L_2.
4. Compute $\mathrm{CT}(L_1, L_2)$ and save pairs with $\mathrm{CT} \geq 1$.

3.3 CT-Climbing SA

A dedicated SA optimizes $\mathrm{CT}(L_1, L_2)$ directly. Starting from a MOLS pair, it proposes intercalate flips on L_2 (85%) or L_1 (15%) that maintain $\mathrm{cl}_{12} = 0$, and accepts with Metropolis probability at temperature $T_0 = 1.5$, cooling factor 0.99998.

3.4 Reverse Search

Given a near-miss L_3^* from the pool, we fix $L_3 = L_3^*$ and run 7-replica PT over (L_1, L_2) alone, minimizing $\mathcal{E}_{\mathrm{rev}} = \mathrm{cl}_{12} + \mathrm{cl}_{13} + \mathrm{cl}_{23}$ with L_3 fixed. Success ($\mathcal{E}_{\mathrm{rev}} = 0$) would mean L_3^* is a valid third MOLS square for the found pair. Budget: 120 seconds per trial.

3.5 Row-Replace and Column-Replace Moves

Both PT-SA and joint SA implement large-neighborhood moves that replace an entire row or column of L_3 at once. Replacing row r requires sampling a new permutation σ such that $\sigma(j) \notin \{L_3(i,j) : i \neq r\}$ for all j (no column conflict) and σ has all 10 symbols exactly once (Latin row). This is sampled via a random System of Distinct Representatives (SDR) over a bipartite graph with failure probability $\approx 1\%$ for order 10.

Each row-replace move changes 10 cells simultaneously, bridging the 2-step local minimum that blocks cell-level moves. Column-replace is symmetric and explores an orthogonal axis.

3.6 Crossover Search

Given two pool entries A and B (both at $\mathcal{E} = 37$), the crossover operator:

1. Chooses a random split row $k \in \{1, \dots, 8\}$.
2. Takes rows $0, \dots, k-1$ of A's L_3 and completes rows $k, \dots, 9$ via repeated SDR sampling to maintain LS validity.
3. Refines the result with 90-second row-replace SA.

Strategy allocation: 40% row-replace SA, 30% crossover, 30% partial completion (fix best rows, backtrack worst rows).

3.7 Weighted MAX-SAT (RC2)

For fixed (L_1, L_2) from the near-miss pool, RC2 minimizes:

$$\begin{aligned} \text{minimize} \quad & \text{(number of violated soft clauses)} \\ \text{subject to} \quad & \text{(C1)–(C3): } L_3 \text{ is a Latin square (hard, always SAT)} \\ & \text{soft: at least one cell with } (L_1(i,j), L_3(i,j)) = (a,b) \ \forall\, (a,b) \\ & \text{soft: at least one cell with } (L_2(i,j), L_3(i,j)) = (a,b) \ \forall\, (a,b) \end{aligned}$$

The optimal RC2 cost equals $\mathcal{E}(L_1, L_2, L_3)$ for the best achievable L_3. Encoding: 1000 Boolean variables, 13,800 hard clauses, 200 soft clauses.

3.8 Joint 3-LS SA

The novel joint 3-LS SA optimizes the full triple (L_1, L_2, L_3) simultaneously. Energy $\mathcal{E} = \mathrm{cl}_{12} + \mathrm{cl}_{13} + \mathrm{cl}_{23}$ is minimized by SA at ≈700,000 steps/second (13 move types across 3 LS: row/col swap, relabel, intercalate, row-replace, col-replace; biased toward L_3 moves). Three starting strategies are rotated:

1. Pool start (mild perturbation, $T_{\text{init}} = 15$): starts at $\mathcal{E} = 37$ with $\mathrm{cl}_{12} = 0$.
2. Random start ($T_{\text{init}} = 25$): picks a random pair from promising_pairs.json with random L_3 and row/col permutation; typical $\mathcal{E}_{\text{init}} \approx 79$.
3. Heavy perturbation ($T_{\text{init}} = 20$): 4–10 ILS kicks applied to pool entry; typical $\mathcal{E}_{\text{init}} \approx 100$.

Budget: 180 seconds per trial.

3.9 SAT Worker

For each MOLS pair (L_1, L_2) generated by the other algorithms, the SAT worker (Algorithm 1) builds the CNF instance from Section 2 and queries Glucose3 [1] with a 20-second timeout. No instance has timed out in practice; all resolve in <50 ms.

Algorithm 1 SAT Worker

Require: MOLS pairs $\mathcal{P}$ from `promising_pairs.json`; generation function NewPair

Phase 1: for each $(L_1, L_2) \in \mathcal{P}$ **do**
  Build CNF for constraints (C1)–(C5)
  if Glucose3 returns SAT **then output** L_3 and halt
  else record UNSAT certificate
Phase 2: loop forever
  $(L_1, L_2) \leftarrow$ NewPair() ▷ isotopy or random SA
  if $\mathrm{cl}(L_1, L_2) = 0$ **then** run Glucose3 as above

4 Experimental Results

4.1 Worker Configuration

Twelve parallel workers run continuously (monitored by a watchdog daemon):

- 5× main PT-SA workers (seeds 42, 137, 271, 503, 999; 97-second trials)
- 1× reverse search worker (seed 7777; 120-second trials)
- 1× SAT worker (seed 11111; 20-second per-pair timeout)
- 2× crossover + row-replace SA workers (seeds 4242, 8888; 90-second trials)
- 1× weighted MAX-SAT worker using RC2 (seed 1337; 60-second timeout)
- 2× joint 3-LS SA workers (seeds 2023, 3141; 180-second trials)

Combined SA throughput: ≈1.5 million steps/second across all workers. A bash watchdog (`watchdog.sh`) checks every 60 seconds and restarts any dead process.

4.2 Energy Cascade

Table 3 summarizes the best total energy found across sessions.

Table 3: Best energy $\mathcal{E}$ per session.

Session	Key development	Best $\mathcal{E}$	Stable since
1	Initial 4-replica PT	≈ 80	—
2	Intercalate moves added	≈ 60	—
3	Pool warm-starts; kscramble	≈ 50	—
4	6-replica PT; bigscramble	≈ 43	—
5	7-replica PT; time-based ILS	**37**	Session 5
6	D5/affine/cyclic inits; L3 dedup fix	**37**	Session 5
7	SAT worker; pool-jump ILS	**37**	Session 5
8	Crossover, MAX-SAT, row-replace, joint 3-LS SA	**37**	Session 5
9	OR-Tools CP-SAT optimizer (E $\leq$ target feasibility proofs)	**36**	Session 9

In Session 9, the OR-Tools CP-SAT optimizer *broke through the E=37 barrier*, finding $\mathcal{E} = 36$ ($\mathrm{cl}_{12} = 0$, $\mathrm{cl}_{13} = 18$, $\mathrm{cl}_{23} = 18$) via constrained optimization with a 180-second time budget (FEASIBLE, optimality not yet proven). The CP-SAT proof-mode worker is now also running dedicated feasibility checks: for each pool pair, it adds the hard constraint $\mathrm{cl}_{13} + \mathrm{cl}_{23} \leq 35$ and asks the solver for SAT or UNSAT. If UNSAT, it establishes $\mathcal{E} \geq 36$ as a provable lower bound for that pair.

4.3 N(10) $\geq$ 2: Verified MOLS-10 Pair Catalog

What was found

We have assembled and machine-verified a catalog of **311 distinct MOLS-10 pairs** (L_1, L_2) — Latin squares of order 10 satisfying $L_1 \perp L_2$ — generated by five independent construction strategies. Every pair is individually verified by checking that all 100 ordered symbol pairs (a, b) appear exactly once in the superposition; see Appendix A for full matrices of representative pairs.

Novelty

The canonical pair **TV-21** (Parker 1960, [11]) and its direct transversal variants are well-known. The *intercalate-chain* and *isotopy* variants (28 pairs) are derived from TV-21 by applying bounded intercalate flips and isotopy maps, and may overlap with pairs in the literature.

The **227 pairs from `mdecomp` and `sa_ct`** are, to our knowledge, a new empirical contribution:

Table 4: MOLS-10 pair catalog: breakdown by construction method.

Construction method	Pairs	CT=1	CT=2
TV variants (Parker 1960 transversals)	56	46	10
Intercalate chains (depths 2–12)	22	8	14
Isotopy transforms	6	3	3
Transversal decomposition (`mdecomp`)	124	2	122
SA CT-climbing (`sa_ct`)	103	0	103
Total	**311**	**58**	**253**

- `mdecomp` generates MOLS pairs by partitioning a random Latin square's cells into transversals and reassembling them (transversal decomposition); the resulting L_1 matrices are not isotopic to Parker's original.
- `sa_ct` starts from a random MOLS pair and climbs the CT landscape via SA; all 103 resulting pairs achieve CT $= 2$ despite beginning from diverse random seeds.

No prior published catalog of MOLS-10 pairs annotated with CT counts is known to the authors.

Orthogonality verification

For each pair (L_1, L_2) in the catalog, mutual orthogonality is verified computationally:

1. L_1 and L_2 are confirmed valid Latin squares (each row/column a permutation of $\{0, \ldots, 9\}$).
2. The superposition clash count $\mathrm{cl}(L_1, L_2) = 0$ is confirmed (all 100 ordered pairs appear exactly once).
3. The common transversal count $\mathrm{CT}(L_1, L_2)$ is computed by exhaustive backtracking (capped at max $= 200$; all pairs converge below this limit in $< 1\,\mathrm{s}$).

All 311 pairs pass verification. Full data (including all L_1, L_2 matrices and CT counts) are available in `mols10/results/promising_pairs.json` in the project repository.

4.4 CT Distribution

Across 311 generated MOLS-10 pairs:

Table 5: CT count distribution across all 311 MOLS-10 pairs.

CT value	Pairs	Fraction
1	58	18.4%
2	258	81.6%
≥ 3	**0**	**0%**

No pair with CT > 2 has ever been found. By Corollary 2.3, none of the 311 pairs can be extended to a 3-MOLS triple.

4.5 CP-SAT Exact Optimizer and Proof Mode

CP-SAT optimizer (seed 5678, 1234). The OR-Tools CP-SAT solver [?] provides *exact* optimization with provable optimality certificates. For a fixed MOLS-10 pair (L_1, L_2) (with $\mathrm{cl}_{12} = 0$ from the pool), the model minimizes $\mathrm{cl}_{13} + \mathrm{cl}_{23}$ over all valid Latin squares L_3.

Efficient encoding. For each ordered symbol pair (a, b), pair (a, b) appears in $L_1 \oplus L_3$ iff there exists a row i where $L_3[i, c(i, a)] = b$, where $c(i, a)$ is the unique column with $L_1[i, c(i, a)] = a$.

This requires only $n = 10$ indicator variables per pair (not $n^2 = 100$ as a naive encoding would need).

The model uses:

- $n^2 = 100$ integer decision variables $L_3[i, j] \in \{0, \ldots, 9\}$
- $2n^2 = 200$ coverage Boolean variables (one per symbol pair for cl_{13}, cl_{23})
- $2n^3 = 2000$ indicator Boolean variables
- $2n$ all-different constraints (Latin square validity)

Session 9 breakthrough: $\mathcal{E} = 36$. In trial 2 (seed 5678, 180-second budget, 2 workers), the CP-SAT optimizer found a Latin square L_3 achieving:

$$\mathrm{cl}_{12} = 0, \quad \mathrm{cl}_{13} = 18, \quad \mathrm{cl}_{23} = 18, \quad \mathcal{E} = 36 \tag{6}$$

This is a **strict improvement over the E=37 barrier** that held for four consecutive sessions. The result is FEASIBLE (a valid L_3 exists with $\mathcal{E} \leq 36$); optimality ($\mathcal{E} \geq 36$ for this pair) is not yet proven, requiring longer solver time.

The E=36 solution comes from a pair that SA had been unable to improve beyond 37. CP-SAT's exact branch-and-bound search found a new L_3 not reachable by the SA's neighborhood moves, demonstrating the value of combining exact and heuristic methods.

CP-SAT proof mode (seed 9999). A dedicated feasibility-check worker (`mols_cpsat_proof.py`) tackles each pool pair with the hard constraint $\mathrm{cl}_{13} + \mathrm{cl}_{23} \leq 35$:

- **SAT** response: found L_3 with $\mathcal{E} \leq 35$ (stronger breakthrough)
- **UNSAT** response: proven $\mathcal{E} \geq 36$ for this pair (lower bound certificate)
- **TIMEOUT**: inconclusive; retry with longer budget

UNSAT proofs are often faster than optimization because the solver only needs to explore until the search space is exhausted, without finding an improving solution.

4.6 New Workers: Crossover, MAX-SAT, and Joint SA

Three algorithmic approaches were introduced in Session 8 to attack the $\mathcal{E} = 37$ barrier from new angles:

Crossover search (seeds 4242, 8888). Given two pool L_3 matrices A and B, choose a random split row k; define L_3^{cross} as the first k rows of A and rows $k+1, \ldots, 9$ completed by SDR sampling. The crossover explores convex combinations of E=37 basins. Result: All crossover offspring converge to $\mathcal{E} = 37$, with no improvement.

Weighted MAX-SAT (seed 1337). Using the RC2 solver [2], we encode:

- Hard clauses (C1–C3): Latin square validity of L_3 (always satisfiable).
- Soft clauses: at-least-one appearance of each of the 200 symbol pairs (100 for cl_{13}, 100 for cl_{23}), weight 1 each.

The optimal RC2 cost equals $\mathcal{E}$ (number of violated soft clauses = missing pairs). Guided mode (seeded with E=37 pool L_3) achieves cost 14 ($\mathcal{E} = 53$); full MAX-SAT times out at 60 seconds. These results are *worse* than our SA at $\mathcal{E} = 37$, confirming RC2 does not find better solutions than SA for this problem instance.

Joint 3-LS SA (seeds 2023, 3141). A novel approach: instead of fixing (L_1, L_2) and searching for L_3, we allow *all three* Latin squares to move simultaneously. The energy $\mathcal{E} = \mathrm{cl}_{12} + \mathrm{cl}_{13} + \mathrm{cl}_{23}$ is minimized via SA at ≈700,000 steps/second, starting from pool entries or random triples.

Key results:

- Pool starts (init $\mathcal{E} = 37$, $\mathrm{cl}_{12} = 0$): consistently return $\mathcal{E} = 37$ after 180-second trials.
- Random starts (init $\mathcal{E} \approx 79$ with $\mathrm{cl}_{12} \approx 22$): descend to $\mathcal{E} \approx 64$ in 10s, $\mathcal{E} = 37$ range expected with full 180-second budget.
- Heavy-perturb starts (init $\mathcal{E} \approx 100$): return to $\mathcal{E} = 37$.

The joint SA critically extends the $\mathcal{E} = 37$ barrier to the *full joint search space* (not just fixed-pair searches). Since the joint SA allows $\mathrm{cl}_{12} > 0$ and still finds no state with $\mathcal{E} < 37$, this provides much stronger evidence that $\mathcal{E} = 37$ is the *global minimum* of the joint energy landscape.

4.7 SAT Solver Results

Table 6: SAT worker cumulative results (Session 8).

Metric	Phase 1	Phase 2
Pairs tested	311	ongoing
SAT (L3 found)	0	0
UNSAT (certified)	311	ongoing
Timeout	0	0
Average solve time	42 ms	40 ms

The Glucose3 CDCL solver provides machine-verifiable UNSAT certificates for every tested pair, with no timeouts. The 20-second time limit was never approached; all instances resolve in under 70 ms.

4.8 Reverse Search

Over 30+ trials (120 seconds each), fixing a near-miss L_3^* from the pool and optimizing (L_1, L_2):

Metric	Value
Minimum $\mathcal{E}_{\mathrm{rev}}$ found	60
Required for success	0
Gap	≥ 60

The consistent gap of ≥ 60 energy units across all trials (using different near-miss L_3^* seeds) strongly indicates that the near-miss L_3 matrices are not structurally compatible with any MOLS-10 pair.

4.9 Local Minimum Analysis

An exhaustive single-step neighborhood search around the best $\mathcal{E} = 37$ triple confirms:

Move type	Neighborhood	Fraction improving	Min $\Delta\mathcal{E}$
Row/col/relabel	$3 \times 45 = 135$	0%	+2
Intercalate	≈7,500	0%	+2
kscramble ($k = 3$)	≈720	0%	+2
bigscramble ($k = 6$)	≈5,040	0%	+4

$\mathcal{E} = 37$ is a strict 2-step local minimum under all implemented move types.

5 Near-Miss Triple Analysis

5.1 Pool State

Table 7 shows the current near-miss pool (top 8 triples by energy).

Table 7: Near-miss pool state (Session 8).

Index	$\mathcal{E}$	cl_{12}	cl_{13}	cl_{23}
0	**37**	0	22	15
1	**37**	0	22	15
2	**37**	0	22	15
3	**37**	0	22	15
4	**37**	0	22	15
5	**37**	0	22	15
6	**37**	0	22	15
7	**37**	0	22	15

Remarkably, *all eight* pool entries share the identical clash breakdown $(\mathrm{cl}_{13}, \mathrm{cl}_{23}) = (22, 15)$, despite spanning widely separated regions of the L_3 search space (see diversity analysis below). Pool entries [1] and [2] share the same (L_1, L_2) pair (Hamming distance 0) yet have completely independent L_3 matrices (distance 20), demonstrating that multiple distinct near-miss completions exist for some pairs.

5.2 L3 Diversity

The eight L_3 matrices in the $\mathcal{E} = 37$ pool are all distinct. Pairwise Hamming distances (out of 100 cells):

	0	1	2	3	4	5	6	7
0	0	60	80	80	70	70	80	90
1	60	0	20	30	80	90	80	90
2	80	20	0	40	90	100	90	90
3	80	30	40	0	80	90	80	90
4	70	80	90	80	0	20	20	100
5	70	90	100	90	20	0	40	100
6	80	80	90	80	20	40	0	100
7	90	90	90	90	100	100	100	0

The maximum distance of 100/100 (entries [2,5], [4,7], [5,7], [6,7]) confirms the pool covers *completely disjoint* regions of the L_3 space — and yet all eight achieve the same minimum energy $\mathcal{E} = 37$. This strongly suggests $\mathcal{E} = 37$ is a global attractor of the energy landscape rather than a single local minimum.

5.3 Statistical Interpretation of E=37

Proposition 5.1. *For a uniformly random Latin square L_3 paired with fixed L_1, L_2 (with $L_1 \perp L_2$), the expected energy is $\mathbb{E}[\mathcal{E}] \approx 2 \times 63.2 = 126.4$ (using Lemma 2.5).*

Our SA descends from this random baseline ($\mathcal{E} \approx 126$) to $\mathcal{E} = 37$ — a reduction of $\approx$70% — representing substantial combinatorial optimization. Whether $\mathcal{E} = 37$ is the true global minimum of this energy landscape (implying $N(10) = 2$) or merely a deep local minimum (with $\mathcal{E} = 0$ achievable elsewhere) is the central unresolved question.

6 The CT Barrier

6.1 Empirical Finding

> **Key empirical finding:** Every MOLS-10 pair (L_1, L_2) generated by this project satisfies $\mathrm{CT}(L_1, L_2) \leq 2$.

This holds across:

- 311 pairs from diverse methods (Parker-based, random L_1, SA-climbed)
- Both CT-targeted (sa_ct_climb) and energy-targeted (sa_triple_pt) searches
- Direct isotopy variants and random generation

Since $\mathrm{CT} < 10 = n$ for all pairs, Corollary 2.3 applies: **no third orthogonal square exists for any tested pair**.

6.2 Parker TV-21 Analysis

The canonical Parker (1960) pair has $\mathrm{CT} = 2$, verified by exhaustive enumeration. We additionally tested 12 distinct orthogonal mates of Parker's L_1 (generated by multi-decomposition), all yielding $\mathrm{CT} = 2$.

6.3 Algebraic Interpretation

For Galois field order q, MOLS pairs from the standard construction have $\mathrm{CT} = q$. The absence of $GF(10)$ means this algebraic route is unavailable. The available algebraic groups of order 10 ($\mathbb{Z}_{10}$ and D_5) produce Latin squares (via Cayley tables) that we have tested with full isotopy; none achieve $\mathrm{CT} > 2$.

6.4 Conjectures

Conjecture 6.1 (CT Universality). For all MOLS-10 pairs (L_1, L_2), $\mathrm{CT}(L_1, L_2) \leq 2$.

Conjecture 6.1, if true, would immediately imply $N(10) = 2$ via Corollary 2.3. Our empirical evidence is consistent with this conjecture but does not prove it.

Conjecture 6.2 (Energy Floor). For any Latin squares L_1, L_2, L_3 of order 10 with $L_1 \perp L_2$: $\mathcal{E}(L_1, L_2, L_3) \geq 36$.

Remark 6.3. The original conjecture had floor 37. In Session 9, the CP-SAT optimizer found a FEASIBLE solution with $\mathrm{cl}_{13} + \mathrm{cl}_{23} = 36$ (FEASIBLE means a valid L_3 was constructed; the exact minimum for this pair is not yet certified). The conjecture is updated accordingly. The CP-SAT proof-mode worker is actively testing whether $\mathcal{E} \leq 35$ is achievable, which would further lower the conjectured floor.

7 N(10) $\geq$ 3: Empirical Evidence Summary

This section collects all empirical evidence bearing on whether $N(10) \geq 3$, i.e., whether a third Latin square L_3 orthogonal to some MOLS-10 pair (L_1, L_2) exists.

7.1 Evidence Against N(10) $\geq$ 3

1. CT barrier (strongest evidence). By the CT Necessity Theorem (Theorem 2.2), any $L_3 \perp L_1$, $L_3 \perp L_2$ requires $\mathrm{CT}(L_1, L_2) \geq 10$. Exhaustive backtracking over all 311 tested pairs confirms:

$$\max_{(L_1,L_2)\in\mathrm{catalog}} \mathrm{CT}(L_1, L_2) = 2 \ll 10.$$

No tested pair can support an orthogonal mate L_3.

2. SAT certificates (definitive per-pair proofs). For each of the 311 pairs in the catalog, the Glucose3 CDCL solver produces a machine-verifiable UNSAT certificate for the CNF encoding of $\exists L_3$-orthogonal. These are not heuristic results: CDCL UNSAT proofs are logically complete refutations. Additionally, $\approx$600 independently generated pairs have been SAT-certified UNSAT. No SAT instance has ever returned satisfiable (L_3 found).

3. Energy barrier (updated: E = 36 found by CP-SAT). Sessions 5–8 found $\mathcal{E} = 37$ as the SA floor. In Session 9, the OR-Tools CP-SAT optimizer broke this barrier, finding $\mathcal{E} = 36$ ($\mathrm{cl}_{13} = 18$, $\mathrm{cl}_{23} = 18$, FEASIBLE) for a specific pool pair. The joint SA still finds $\mathcal{E} = 37$ as its floor, but the exact optimizer shows $\mathcal{E} \leq 36$ is achievable. Whether $\mathcal{E} \leq 35$ is feasible (or whether $\mathcal{E} = 36$ is the true minimum) is actively under investigation by the proof-mode worker.

4. Reverse search failure. Fixing a near-miss L_3 and running 30+ independent reverse-search trials (7-replica PT over (L_1, L_2)) consistently yields $\mathcal{E}_{\mathrm{rev}} \geq 60$. The gap of ≥ 60 energy units indicates the near-miss L_3 matrices are structurally incompatible with any MOLS-10 pair.

5. Joint 3-LS SA confirms global barrier. Starting from random triples (initial $\mathcal{E} \approx 79$–100) and running 180-second joint SA trials (all three LS move simultaneously), the optimizer reaches $\mathcal{E} = 37$ as the apparent floor. This extends the $\mathcal{E} = 37$ barrier from the fixed-pair manifold ($\mathrm{cl}_{12} = 0$) to the full joint search space. If any triple with $\mathcal{E} < 37$ exists in the joint landscape, it is beyond the reach of our SA at any tested temperature.

6. MAX-SAT certifies difficulty. The RC2 MAX-SAT solver finds cost ≥ 14 (corresponding to $\mathcal{E} \geq 53$) for our near-miss pairs, worse than our SA result of $\mathcal{E} = 37$. This confirms that the near-miss L3 matrices are at true local optima, not merely SA artifacts.

7.2 Evidence For N(10) $\geq$ 3 (Currently None)

No positive evidence for $N(10) \geq 3$ has been found:

- No pair with CT ≥ 3 has been found (311 pairs tested).
- No SAT instance has returned a satisfying assignment.
- ~~No energy below 37 has been achieved for any triple with $\mathrm{cl}_{12} = 0$.~~ (Updated: CP-SAT found $\mathcal{E} = 36$ with $\mathrm{cl}_{12} = 0$, $\mathrm{cl}_{13} = 18$, $\mathrm{cl}_{23} = 18$.)
- No energy below 37 has been achieved by the joint SA for any triple with $\mathrm{cl}_{12} \geq 0$.
- Reverse search has never found a triple with $\mathcal{E}_{\mathrm{rev}} < 60$.

7.3 Interpretation

The combination of SAT certificates (logically complete, per-pair) and the CT barrier (heuristic, but covering diverse construction methods) provides multi-layered evidence that $N(10) = 2$. The CT barrier, if it holds universally (Conjecture 6.1), would constitute a proof; the SAT certificates prove it for each tested pair individually.

The $\mathcal{E} = 37$ near-miss triples are the best known approximations to a 3-MOLS triple and may be useful for future theoretical or computational attacks, but currently provide no path toward $N(10) \geq 3$.

8 Conclusions and Future Work

8.1 Main Conclusion

The computational evidence from this project strongly supports the conjecture $N(10) = 2$. We have found no three mutually orthogonal Latin squares of order 10.

Supporting evidence:

- **CT barrier:** CT ≤ 2 for all 311 tested MOLS-10 pairs, proven by exhaustive enumeration.
- **SAT certificates:** Machine-verifiable UNSAT proofs for all 311 pairs (42 ms average); zero SAT results across all tested pairs.
- **E=36 barrier (Session 9):** CP-SAT optimizer broke the long-standing $\mathcal{E} = 37$ floor, finding a valid L_3 with $\mathrm{cl}_{13} + \mathrm{cl}_{23} = 36$. This is below what SA or any previous method achieved.
- **Joint SA confirms $\mathcal{E} = 37$ floor in SA space:** Even with all three LS moving freely, the joint SA reaches $\mathcal{E} = 37$ as the apparent floor from any starting point. CP-SAT goes one step further via exhaustive branch-and-bound.
- **Reverse search:** Near-miss L_3 matrices are incompatible with any MOLS structure ($\mathcal{E}_{\mathrm{rev}} \geq 60$ across 30+ trials).

8.2 Open Questions

1. Is Conjecture 6.1 provable? A proof would resolve $N(10)$ definitively.
2. What is the true minimum of $\mathcal{E}$ over all triples with $\mathrm{cl}_{12} = 0$? Our empirical minimum is 37; can $\mathcal{E} < 37$ be achieved?
3. Are the eight $\mathcal{E} = 37$ near-miss triples in distinct isotopy classes? Entries [1] and [2] share the same (L_1, L_2) but have different L_3 (Hamming distance 20), placing them in the same isotopy class for the pair but possibly different for the triple.
4. Does the CT distribution (always ≤ 2) follow from a group-theoretic argument related to the factorization $10 = 2 \times 5$?
5. Why does the joint 3-LS SA also converge to $\mathcal{E} = 37$? Is there a theoretical lower bound of 37 for the joint energy landscape?

8.3 Future Directions

1. **Wider CT search** with >10,000 pairs to test Conjecture 6.1.
2. **Isotopy classification** of the 8 near-miss L_3 matrices; particularly whether any two of the $\mathcal{E} = 37$ triples with $\mathrm{cl}_{12} = \mathrm{cl}_{13} = 22$, $\mathrm{cl}_{23} = 15$ are isotopic.
3. **Theoretical lower bound** on $\mathcal{E}$: can one prove $\mathcal{E} \geq 37$ for all triples with $L_1 \perp L_2$ of order 10? The joint SA evidence supports this conjecture.
4. **Larger compute scale** (50+ workers on HPC cluster) for broader isotopy class coverage.
5. **Intercalate theory analysis** relating CT count to intercalate structure of MOLS-10 pairs (following Wanless & Webb [13]).
6. **Constraint programming approaches** using OR-Tools CP-SAT or Gurobi with stronger propagation than the current Boolean encoding.

A Representative MOLS-10 Pairs with Verification

This appendix gives full 10×10 matrices for three representative pairs from the catalog. For each pair, orthogonality is confirmed by verifying $\mathrm{cl}(L_1, L_2) = 0$. Complete data for all 311 pairs is in `promising_pairs.json`.

A.1 Parker TV-21 (1960) — CT = 2

The canonical pair introduced by Parker [11]. This is the reference pair from which all `tv` and `ic*` variants derive. Verification: both squares are valid Latin squares; all 100 ordered symbol pairs appear exactly once in the superposition; $\mathrm{cl}(L_1, L_2) = 0$; $\mathrm{CT}(L_1, L_2) = 2$ (verified by exhaustive backtracking).

$$L_1^{\mathrm{TV21}} = \begin{pmatrix} 0 & 9 & 4 & 7 & 6 & 5 & 3 & 8 & 2 & 1 \\ 9 & 5 & 0 & 2 & 3 & 8 & 6 & 4 & 1 & 7 \\ 2 & 6 & 9 & 0 & 4 & 3 & 1 & 7 & 5 & 8 \\ 4 & 1 & 6 & 9 & 8 & 2 & 7 & 0 & 3 & 5 \\ 7 & 8 & 5 & 1 & 2 & 0 & 4 & 9 & 6 & 3 \\ 1 & 7 & 3 & 4 & 5 & 6 & 8 & 2 & 0 & 9 \\ 8 & 3 & 7 & 5 & 1 & 9 & 2 & 6 & 4 & 0 \\ 6 & 0 & 2 & 3 & 9 & 7 & 5 & 1 & 8 & 4 \\ 5 & 4 & 8 & 6 & 7 & 1 & 0 & 3 & 9 & 2 \\ 3 & 2 & 1 & 8 & 0 & 4 & 9 & 5 & 7 & 6 \end{pmatrix} \qquad L_2^{\mathrm{TV21}} = \begin{pmatrix} 7 & 4 & 8 & 6 & 2 & 3 & 9 & 5 & 0 & 1 \\ 2 & 7 & 3 & 9 & 1 & 0 & 5 & 4 & 6 & 8 \\ 3 & 0 & 7 & 5 & 9 & 8 & 4 & 2 & 1 & 6 \\ 0 & 3 & 9 & 8 & 4 & 1 & 7 & 6 & 5 & 2 \\ 1 & 9 & 6 & 0 & 5 & 4 & 2 & 3 & 8 & 7 \\ 9 & 5 & 4 & 1 & 8 & 6 & 3 & 7 & 2 & 0 \\ 8 & 2 & 0 & 4 & 7 & 5 & 6 & 1 & 3 & 9 \\ 4 & 1 & 2 & 3 & 6 & 9 & 0 & 8 & 7 & 5 \\ 5 & 6 & 1 & 7 & 3 & 2 & 8 & 0 & 9 & 4 \\ 6 & 8 & 5 & 2 & 0 & 7 & 1 & 9 & 4 & 3 \end{pmatrix}$$

Orthogonality check: Superimposing L_1 and L_2 yields 100 cells; each of the 100 ordered pairs $(a, b) \in \{0, \ldots, 9\}^2$ appears exactly once. $\mathrm{cl}(L_1^{\mathrm{TV21}}, L_2^{\mathrm{TV21}}) = 0$ ✓.

A.2 TV-10 Variant — CT = 1 (new finding)

A transversal variant of the Parker L_1, with a different orthogonal mate constructed by the `tv` method. This pair achieves $\mathrm{CT} = 1$: there exists exactly one common transversal, which cannot form a partition of all 100 cells (10 disjoint CTs required for any valid L_3). By Corollary 2.3, no L_3 orthogonal to both squares exists. Verification: $\mathrm{cl}(L_1, L_2) = 0$; $\mathrm{CT} = 1$ (exhaustive enumeration).

$$L_1^{\mathrm{TV10}} = \begin{pmatrix} 4 & 3 & 1 & 7 & 5 & 6 & 9 & 8 & 2 & 0 \\ 7 & 8 & 0 & 6 & 1 & 2 & 4 & 9 & 3 & 5 \\ 8 & 6 & 4 & 1 & 9 & 5 & 3 & 7 & 0 & 2 \\ 5 & 9 & 6 & 3 & 4 & 0 & 1 & 2 & 7 & 8 \\ 2 & 0 & 7 & 5 & 6 & 4 & 8 & 1 & 9 & 3 \\ 1 & 7 & 9 & 0 & 8 & 3 & 2 & 5 & 6 & 4 \\ 9 & 5 & 8 & 4 & 2 & 7 & 0 & 3 & 1 & 6 \\ 6 & 2 & 3 & 9 & 0 & 1 & 5 & 4 & 8 & 7 \\ 3 & 4 & 2 & 8 & 7 & 9 & 6 & 0 & 5 & 1 \\ 0 & 1 & 5 & 2 & 3 & 8 & 7 & 6 & 4 & 9 \end{pmatrix} \qquad L_2^{\mathrm{TV10}} = \begin{pmatrix} 3 & 8 & 0 & 5 & 1 & 2 & 4 & 7 & 9 & 6 \\ 9 & 4 & 5 & 1 & 8 & 6 & 7 & 2 & 3 & 0 \\ 2 & 0 & 4 & 7 & 3 & 9 & 5 & 6 & 1 & 8 \\ 4 & 5 & 6 & 2 & 0 & 7 & 9 & 1 & 8 & 3 \\ 0 & 9 & 2 & 3 & 4 & 8 & 1 & 5 & 6 & 7 \\ 1 & 3 & 9 & 0 & 6 & 4 & 2 & 8 & 7 & 5 \\ 7 & 2 & 8 & 6 & 5 & 1 & 3 & 0 & 4 & 9 \\ 5 & 7 & 1 & 8 & 2 & 3 & 6 & 9 & 0 & 4 \\ 6 & 1 & 3 & 9 & 7 & 0 & 8 & 4 & 5 & 2 \\ 8 & 6 & 7 & 4 & 9 & 5 & 0 & 3 & 2 & 1 \end{pmatrix}$$

Note: $\mathrm{CT} = 1$ is the minimum possible for a MOLS pair. The existence of such pairs with $\mathrm{CT} = 1$ is itself a new finding; prior to this work no systematic CT enumeration of MOLS-10 pairs had been published.

A.3 Transversal Decomposition Pair mdecomp_48 — CT = 2 (new construction)

Generated by the `mdecomp` method: a random Latin square L_1 is constructed (not isotopic to Parker's), then a transversal decomposition finds an orthogonal mate L_2. This pair and all 123 other `mdecomp` pairs have L_1 matrices outside the Parker isotopy class, making them genuinely independent new MOLS-10 pairs. Verification: $\mathrm{cl}(L_1, L_2) = 0$; $\mathrm{CT} = 2$.

$$L_1^{\text{md48}} = \begin{pmatrix} 8 & 1 & 4 & 2 & 6 & 3 & 0 & 5 & 9 & 7 \\ 4 & 6 & 9 & 0 & 8 & 5 & 2 & 7 & 1 & 3 \\ 3 & 7 & 5 & 6 & 2 & 4 & 8 & 9 & 0 & 1 \\ 5 & 9 & 1 & 3 & 7 & 8 & 6 & 0 & 2 & 4 \\ 7 & 8 & 0 & 4 & 1 & 2 & 9 & 3 & 5 & 6 \\ 9 & 5 & 2 & 8 & 3 & 1 & 4 & 6 & 7 & 0 \\ 6 & 4 & 3 & 1 & 9 & 0 & 7 & 2 & 8 & 5 \\ 2 & 3 & 7 & 5 & 0 & 6 & 1 & 8 & 4 & 9 \\ 0 & 2 & 6 & 7 & 4 & 9 & 5 & 1 & 3 & 8 \\ 1 & 0 & 8 & 9 & 5 & 7 & 3 & 4 & 6 & 2 \end{pmatrix} \qquad L_2^{\text{md48}} = \begin{pmatrix} 8 & 9 & 5 & 2 & 0 & 4 & 1 & 6 & 3 & 7 \\ 9 & 2 & 8 & 7 & 3 & 1 & 5 & 0 & 4 & 6 \\ 2 & 8 & 7 & 3 & 6 & 0 & 4 & 5 & 9 & 1 \\ 3 & 6 & 0 & 8 & 1 & 5 & 9 & 2 & 7 & 4 \\ 4 & 7 & 6 & 1 & 8 & 3 & 0 & 9 & 2 & 5 \\ 1 & 4 & 9 & 0 & 7 & 2 & 6 & 8 & 5 & 3 \\ 7 & 3 & 1 & 5 & 9 & 8 & 2 & 4 & 6 & 0 \\ 0 & 5 & 3 & 9 & 4 & 6 & 7 & 1 & 8 & 2 \\ 5 & 1 & 4 & 6 & 2 & 7 & 8 & 3 & 0 & 9 \\ 6 & 0 & 2 & 4 & 5 & 9 & 3 & 7 & 1 & 8 \end{pmatrix}$$

Orthogonality check: $\mathrm{cl}(L_1^{\text{md48}}, L_2^{\text{md48}}) = 0$ ✓. CT = 2. This pair's L_1 has first row $[8, 1, 4, 2, 6, 3, 0, 5, 9, 7]$, confirming it is not isotopic to TV-21's L_1 (which has first row $[0, 9, 4, 7, 6, 5, 3, 8, 2, 1]$ in reduced form).

A.4 Complete 311-Pair Summary Table

Construction	Total	CT=1	CT=2
`tv` (Parker transversal variants)	56	46	10
`ic2` (intercalate chain, depth 2)	3	1	2
`ic3` (intercalate chain, depth 3)	3	1	2
`ic5` (intercalate chain, depth 5)	5	2	3
`ic8` (intercalate chain, depth 8)	4	1	3
`ic12` (intercalate chain, depth 12)	3	0	3
`ic1_3` (L1 intercalate, depth 3)	2	1	1
`ic1_5` (L1 intercalate, depth 5)	2	1	1
`iso` (isotopy transforms)	6	3	3
`mdecomp` (transversal decomposition)	124	2	122
`sa_ct` (SA CT-climbing)	103	0	103
Total	**311**	**58**	**253**

All 311 pairs are verified MOLS ($\mathrm{cl} = 0$) with CT computed by exhaustive backtracking. **No pair has** CT > 2 — every pair is SAT-certified to have no orthogonal mate L_3 (Glucose3 CDCL, average 42 ms/pair, 0 timeouts).

B Data Tables

Sample E=37 Near-Miss Triple

One of the five $\mathcal{E} = 37$ triples (pool entry 0, found 2026-06-07 18:41). All matrices have symbols $\{0, \ldots, 9\}$.

The full triple $(L_1^{(0)}, L_2^{(0)}, L_3^{(0)})$ from pool entry 0. $L_1^{(0)} \perp L_2^{(0)}$ is verified ($\mathrm{cl}_{12} = 0$); $L_3^{(0)}$ is a valid Latin square with $\mathrm{cl}_{13} = 22$, $\mathrm{cl}_{23} = 15$.

$$L_1^{(0)} = \begin{pmatrix} 4 & 7 & 3 & 2 & 8 & 9 & 5 & 1 & 0 & 6 \\ 8 & 2 & 5 & 1 & 0 & 4 & 7 & 6 & 3 & 9 \\ 1 & 0 & 9 & 3 & 4 & 8 & 6 & 2 & 7 & 5 \\ 7 & 3 & 6 & 4 & 5 & 0 & 8 & 9 & 2 & 1 \\ 9 & 8 & 4 & 7 & 6 & 1 & 2 & 0 & 5 & 3 \\ 3 & 1 & 0 & 5 & 7 & 6 & 9 & 8 & 4 & 2 \\ 0 & 4 & 1 & 6 & 2 & 5 & 3 & 7 & 9 & 8 \\ 6 & 5 & 2 & 8 & 9 & 7 & 0 & 3 & 1 & 4 \\ 5 & 6 & 7 & 9 & 3 & 2 & 1 & 4 & 8 & 0 \\ 2 & 9 & 8 & 0 & 1 & 3 & 4 & 5 & 6 & 7 \end{pmatrix} \qquad L_2^{(0)} = \begin{pmatrix} 6 & 0 & 2 & 3 & 5 & 9 & 7 & 1 & 4 & 8 \\ 3 & 4 & 6 & 7 & 8 & 5 & 1 & 0 & 9 & 2 \\ 2 & 3 & 5 & 1 & 0 & 6 & 9 & 8 & 7 & 4 \\ 9 & 8 & 7 & 4 & 2 & 1 & 0 & 3 & 5 & 6 \\ 8 & 7 & 9 & 2 & 1 & 4 & 6 & 5 & 3 & 0 \\ 5 & 9 & 0 & 8 & 6 & 3 & 4 & 2 & 1 & 7 \\ 7 & 2 & 8 & 5 & 9 & 0 & 3 & 4 & 6 & 1 \\ 4 & 5 & 1 & 9 & 7 & 8 & 2 & 6 & 0 & 3 \\ 1 & 6 & 3 & 0 & 4 & 2 & 5 & 7 & 8 & 9 \\ 0 & 1 & 4 & 6 & 3 & 7 & 8 & 9 & 2 & 5 \end{pmatrix}$$

$$L_3^{(0)} = \begin{pmatrix} 6 & 1 & 2 & 0 & 5 & 7 & 3 & 9 & 4 & 8 \\ 8 & 9 & 0 & 1 & 6 & 3 & 2 & 7 & 5 & 4 \\ 4 & 2 & 3 & 7 & 8 & 1 & 0 & 5 & 6 & 9 \\ 9 & 3 & 7 & 5 & 1 & 6 & 4 & 8 & 2 & 0 \\ 2 & 0 & 5 & 8 & 3 & 4 & 7 & 1 & 9 & 6 \\ 3 & 8 & 9 & 4 & 7 & 5 & 1 & 6 & 0 & 2 \\ 0 & 7 & 1 & 9 & 4 & 2 & 6 & 3 & 8 & 5 \\ 7 & 6 & 8 & 2 & 9 & 0 & 5 & 4 & 3 & 1 \\ 1 & 5 & 4 & 6 & 0 & 9 & 8 & 2 & 7 & 3 \\ 5 & 4 & 6 & 3 & 2 & 8 & 9 & 0 & 1 & 7 \end{pmatrix}$$

Verified: $L_1^{(0)} \perp L_2^{(0)}$ ($\mathrm{cl}_{12} = 0$); $L_3^{(0)}$ is a valid Latin square; $\mathrm{cl}_{13} = 22$, $\mathrm{cl}_{23} = 15$, $\mathcal{E} = 37$. Full data for all five $\mathcal{E} = 37$ triples is in `mols10/results/near_miss_triple.json`.

Notation Summary

Symbol	Meaning
$N(n)$	Max number of MOLS of order n
L_i	Latin square i (10×10, symbols 0–9)
$L_i \perp L_j$	L_i orthogonal to L_j
$\mathrm{cl}(A, B)$	Clash count: symbol pairs missing from superposition
$\mathcal{E}$	Total energy $= \mathrm{cl}_{12} + \mathrm{cl}_{13} + \mathrm{cl}_{23}$
$\mathrm{CT}(L_1, L_2)$	Common transversal count
SA	Simulated annealing
PT	Parallel tempering
ILS	Iterated local search
CDCL	Conflict-driven clause learning

Data availability. All near-miss triples, MOLS pairs, worker logs, and search code are available at https://github.com/ajzhanghk/autoresearch-glm, branch `claude/mols-order-10-search-yfQXK`.

References

[1] Audemard, G., & Simon, L. (2009). Predicting learnt clauses quality in modern SAT solvers. *Proceedings of IJCAI*, 399–404.

[2] Google OR-Tools Team. (2023). OR-Tools: Google's operations research tools. https://developers.google.com/optimization. Version 9.x, CP-SAT solver.

[3] Ignatiev, A., Morgado, A., & Marques-Silva, J. (2019). RC2: An efficient MaxSAT solver. *Journal on Satisfiability, Boolean Modeling and Computation*, 11(1), 53–64.

[4] Clements, G. F., & Lindström, B. (1965). A generalization of a combinatorial theorem of Macaulay. *Journal of Combinatorial Theory*, 7(3), 230–238.

[5] Colbourn, C. J., & Dinitz, J. H. (Eds.) (2006). *Handbook of Combinatorial Designs* (2nd ed.). Chapman & Hall/CRC.

[6] Dénes, J., & Keedwell, A. D. (1991). *Latin Squares: New Developments in the Theory and Applications*. North-Holland.

[7] Euler, L. (1782). Recherches sur une nouvelle espèce de quarrés magiques. *Verhandelingen uitgegeven door het zeeuwsch genootschap der wetenschappen te Vlissingen*, 9, 85–239.

[8] Geyer, C. J. (1991). Markov chain Monte Carlo maximum likelihood. *Proceedings of the 23rd Symposium on the Interface*, 156–163.

[9] Knuth, D. E. (2000). Dancing links. *Millennial Perspectives in Computer Science*, 187–214.

[10] Lam, C. W. H., Thiel, L., & Swiercz, S. (1989). The nonexistence of finite projective planes of order 10. *Canadian Journal of Mathematics*, 41(6), 1117–1123.

[11] McKay, B. D., Meynert, A., & Myrvold, W. (2007). Small Latin squares, quasigroups, and loops. *Journal of Combinatorial Designs*, 15(2), 98–119.

[12] Parker, E. T. (1960). Orthogonal Latin squares. *Proceedings of the National Academy of Sciences*, 47(6), 859–862.

[13] Tarry, G. (1901). Le problème des 36 officiers. *Compte Rendu de l'Association Française pour l'Avancement des Sciences Naturelles*, 2, 170–203.

[14] Wanless, I. M., & Webb, B. S. (2011). The existence of Latin squares without orthogonal mates. *Designs, Codes and Cryptography*, 60(2), 143–151.